
Lecture 6: Interferometers and Aperture Synthesis

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The Quest for Angular Resolution

Angular resolution of a single telescope/antenna is $\delta \sim \lambda/D$: it is the smallest angular separation which two point sources can have in order to be recognized as separate objects.

The 30-meter aperture of a e.g., IRAM antenna provides insufficient resolution for modern astronomy. e.g., 20 arcsecond at 100 GHz, when we want 1 arcsec or much better. The GBT 100 m has only 4 times better resolution than a IRAM antenna at 100 GHz. This issue is more seriously at longer wavelengths.

To increase the resolution, we need to increase D i.e., building a bigger telescope. This trivial solution, however, is not practical. e.g., 1 arcsec resolution at $\lambda = 20$ cm requires a 40 km aperture!!!

Besides, we also need to increase precision of telescope positioning and keep high surface accuracy.

⇒ Technically difficult (perhaps impossible?) to have very large telescope

As this is not practical, we must consider a means of synthesizing the equivalent aperture through combinations of elements. This method, termed “aperture synthesis”, was developed in the 1950s in England and Australia. Martin Ryle (Univ. of Cambridge) earned a Nobel Prize for his contributions.

Submillimeter Array

Aperture synthesis: Replacing a single large telescope by a collection of small telescope “filling” the large one. Technically difficult but feasible. $\Rightarrow$ Interferometers!!



Baseline: Line segment between two antennas

b_{ij} : Baseline length betw. antenna i and j

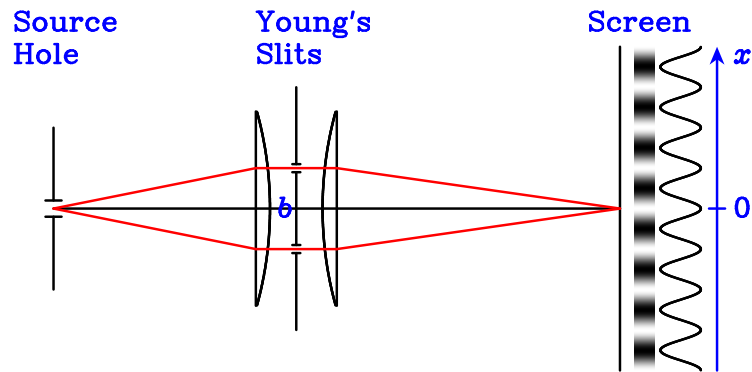
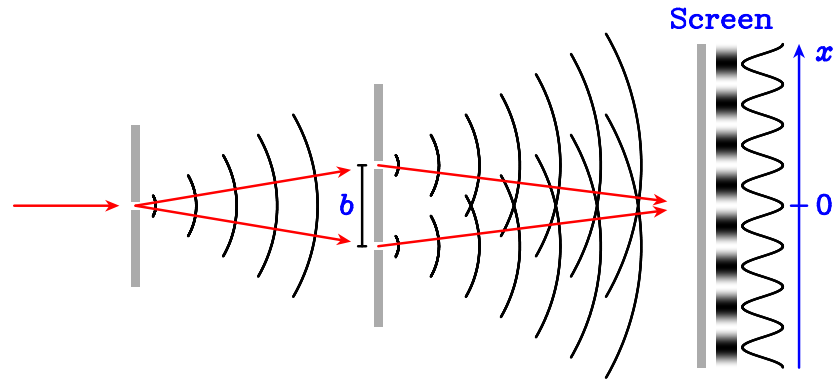
Configuration: Antenna layout (e.g., compact configuration)

D : configuration size (eg. 60 m).

Primary beam: Resolution of one antenna (e.g. 55" at 1.3mm)

Synthesized beam: Resolution of the array (e.g. 3" at 1.3mm)

Young's Double-Slit Experiment

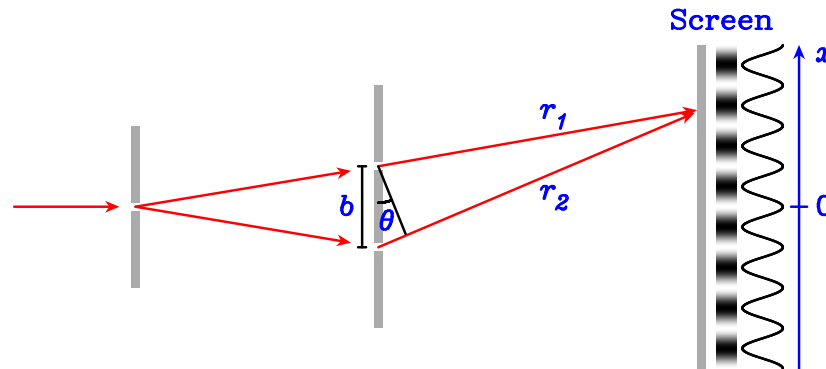


A classic example of **interference effects in light waves in OPTICAL**. Same experiments but with the 2nd one having two lens on either side of the Young's slits.

Lens $\Rightarrow$ Fraunhofer conditions (i.e., Plane waves as if the source were placed at infinity).

Slit width is assumed to be $a \ll \lambda$, so that the slits behave essentially like point sources.

Young's Double-Slit Experiment



Let's assume a Monochromatic source and the rays are in phase when they pass through the slits.

At the screen, let's have $E_{(1)} = E_1 e^{i(kr_1 - \omega t)}$ and $E_{(2)} = E_2 e^{i(kr_2 - \omega t)}$ from the two Young's slits, and thus the total $E = E_{(1)} + E_{(2)}$.

Obtained image of interference: (time averaged) fringes

$$I(x) = \langle E \cdot E^* \rangle = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(x)$$

with $x = k(r_2 - r_1) = bk \sin \theta$, $I_1 = E_1^2$, $I_2 = E_2^2$, $r_2 - r_1 = b \sin \theta$. If $I_1 = I_2 = I$, $I(\theta) = 4I \cos^2\left(\frac{bk \sin \theta}{2}\right)$. Thus $I(x) = 4I$ for constructive interference (bright spots) and $I(x) = 0$ for destructive interference (dark spots). Note intensity $\propto$ square of the amplitude. The condition for constructive interference at the screen is $b \sin \theta = m\lambda$, $m = 0, \pm 1, \pm 2, \dots$

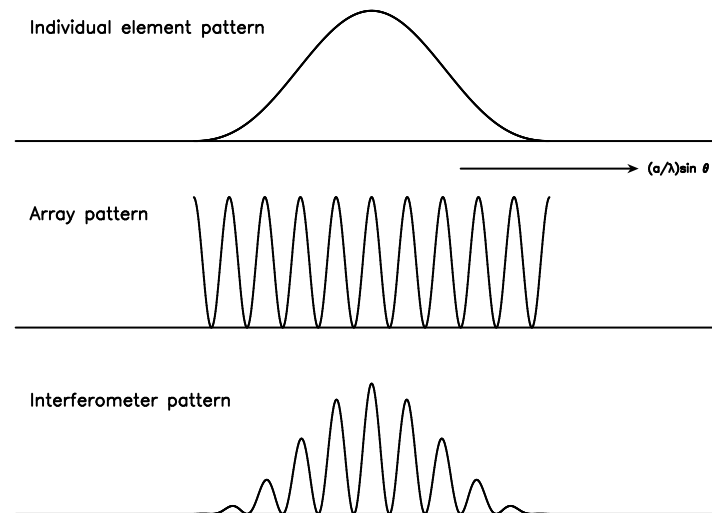
Young's Double-Slit Experiment

Each slit is like a 1-D antenna, with a width of $L_x = a$ and $l = \sin \theta$:

$$P_n(l) = \text{sinc}^2\left(\frac{lL_x}{\lambda}\right) \Rightarrow \text{sinc}^2\left(\frac{a \sin \theta}{\lambda}\right)$$

Note $\int_{-\infty}^{\infty} \text{sinc}^2(x) dx = 1$. With $k = \frac{2\pi}{\lambda}$, then combining the effect of the slit, we have

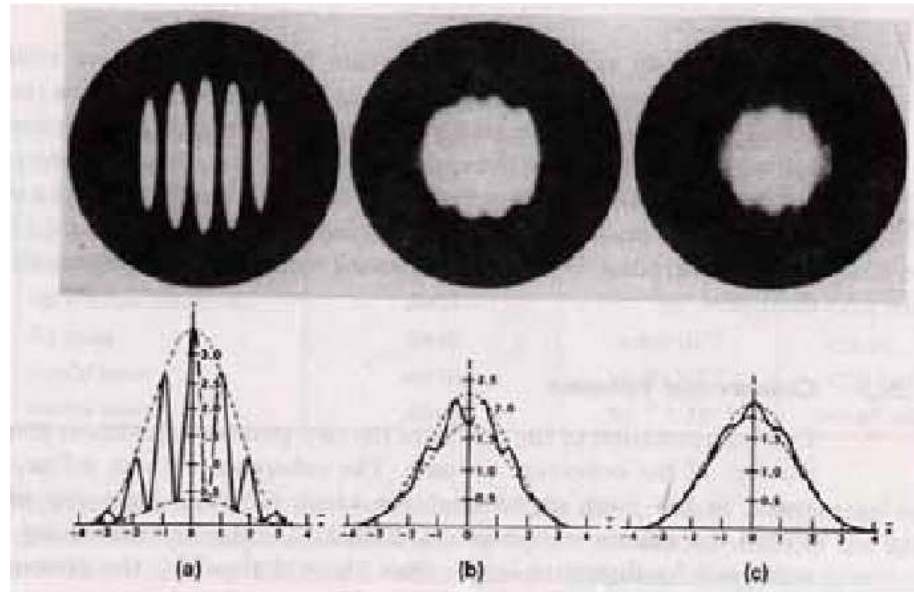
$$I(\theta) = \left(\frac{a}{\lambda}\right)^2 \text{sinc}^2\left(\frac{a \sin \theta}{\lambda}\right) \cdot 4I \cos^2\left(\frac{\pi b \sin \theta}{\lambda}\right)$$



Here $b = 5a$. Slits have the same power pattern as the antenna because of Huygen's Principle.

Young's Double-Slit Experiment

Effect of increasing source size: If the source hole size increases, then the fringes will disappear.



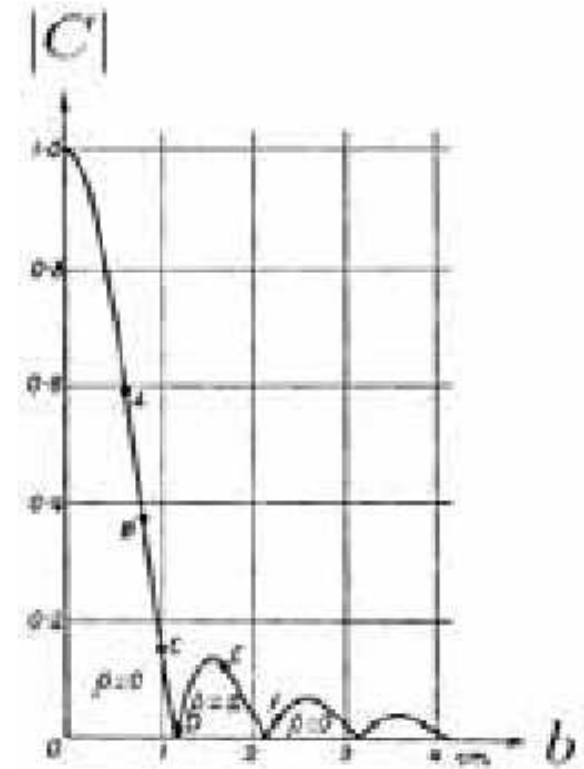
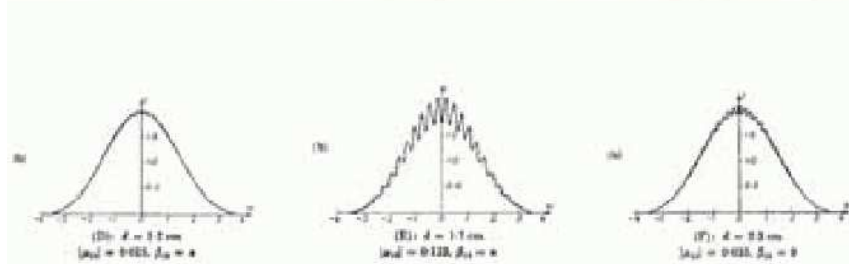
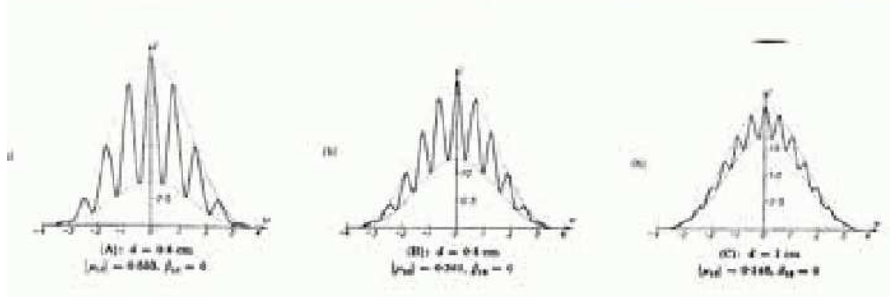
Fringes disappear! $\Rightarrow$ Fringe contrast is linked to the spatial properties of the source.

$$I(x) = I_1 + I_2 + 2\sqrt{I_1 I_2} |C| \cos(x + \phi_C) \quad \text{with} \quad |C| = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

Here $|C|$ a.k.a fringe visibility measured in optical interferometry, a measure of coherency. It is 1 for a coherent source and 0 for not. Here, the coherency is lost as the source size increases, due to the superposition of waves from different (assumed to be incoherent) part of the source. In addition, if the source's "center of brightness" is above (below) the axis betw. the two slits, then the wave from the two slits will have a phase different and the fringe center will shift below (above) $x = 0$.

Young's Double-Slit Experiment

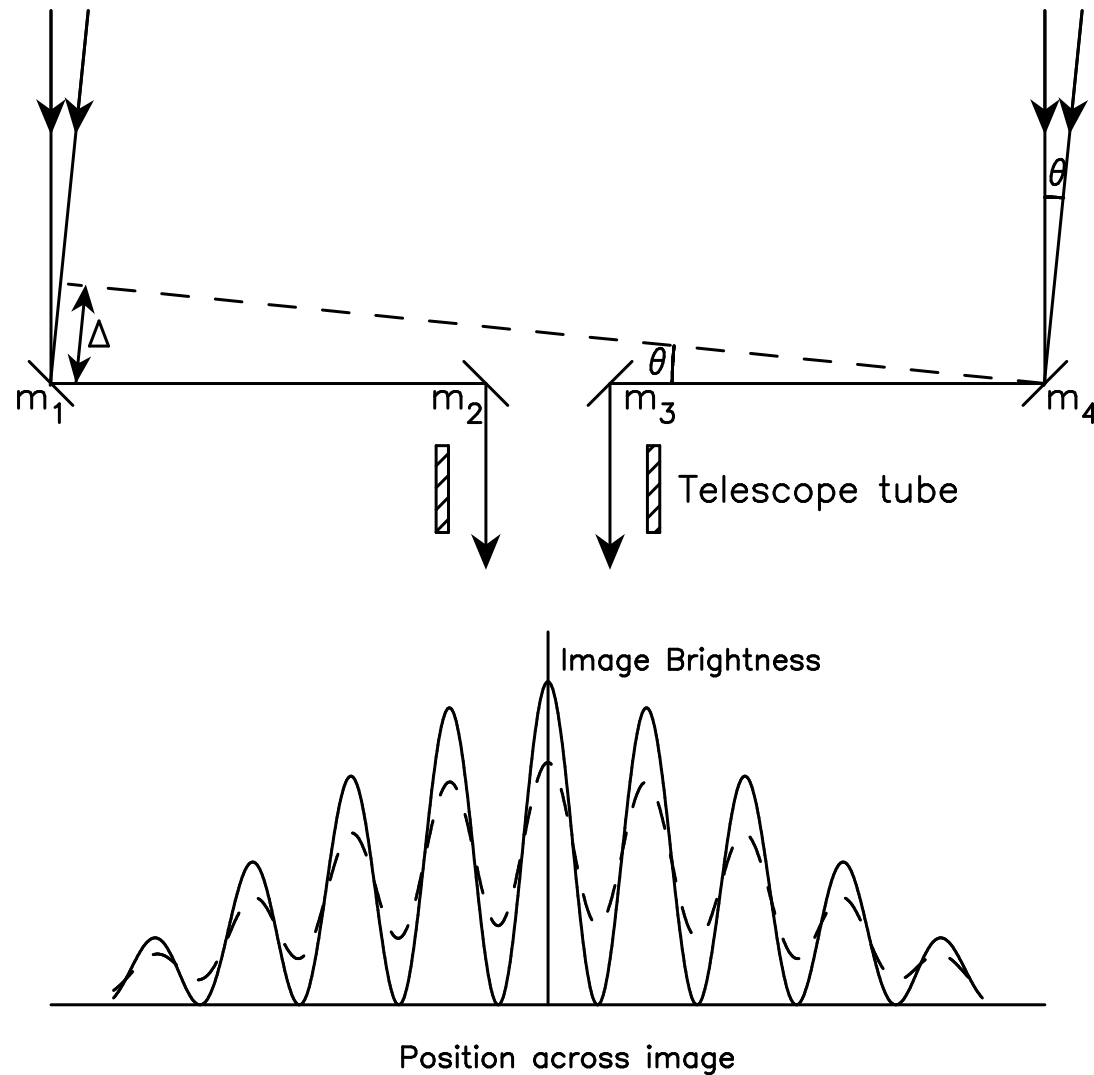
Effect of increasing b : The source is a uniform disk



The Visibility is a 2D-Fourier Transform of the source!

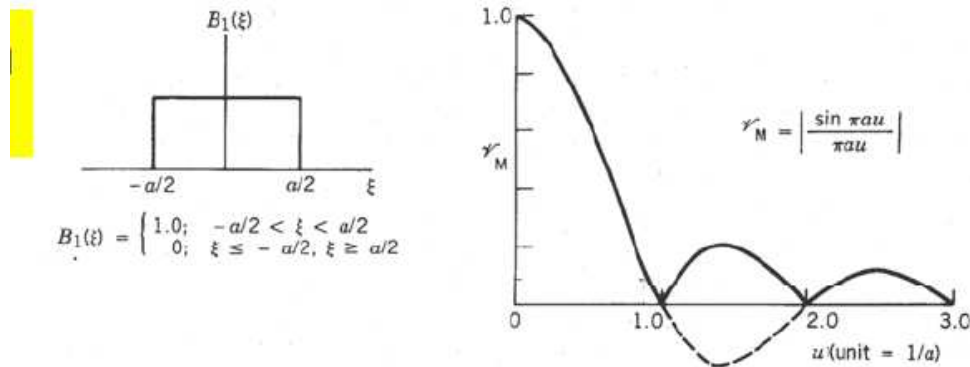
Measure visibility at different $b \Rightarrow$ source properties!

Michelson Interferometer

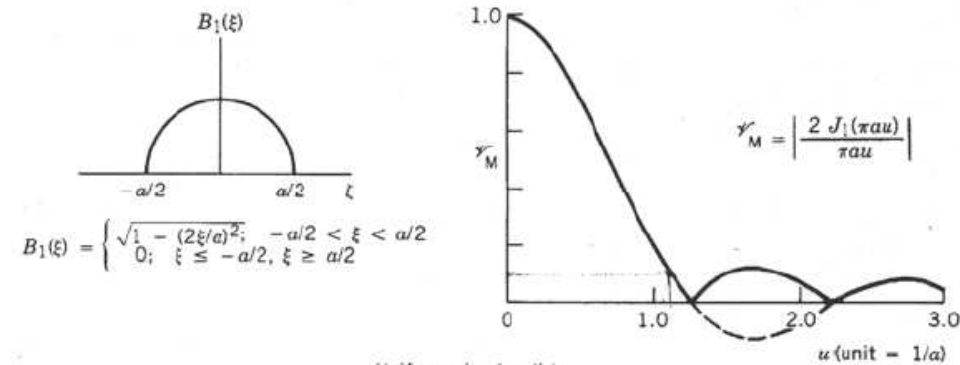


The solid line shows the fringe profiles for an unresolved star ($|C| = 1.0$) and the broken line for a partially resolved star for which ($|C| = 0.5$)

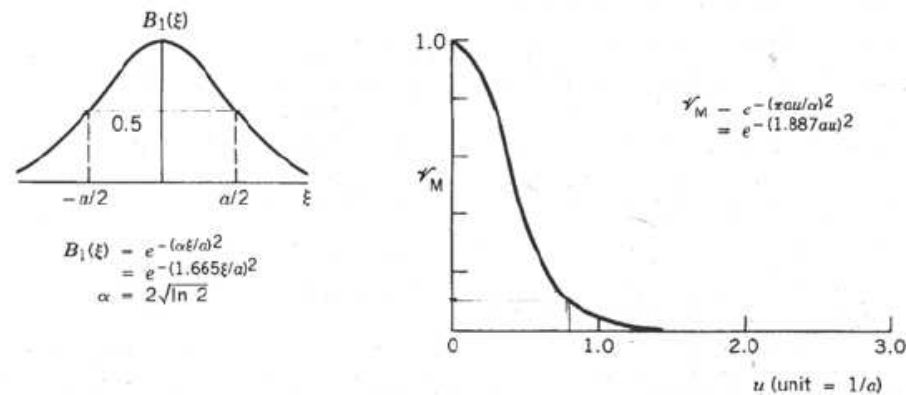
Examples of Visibility Functions



Rectangular model



Uniform circular disk



Gaussian model

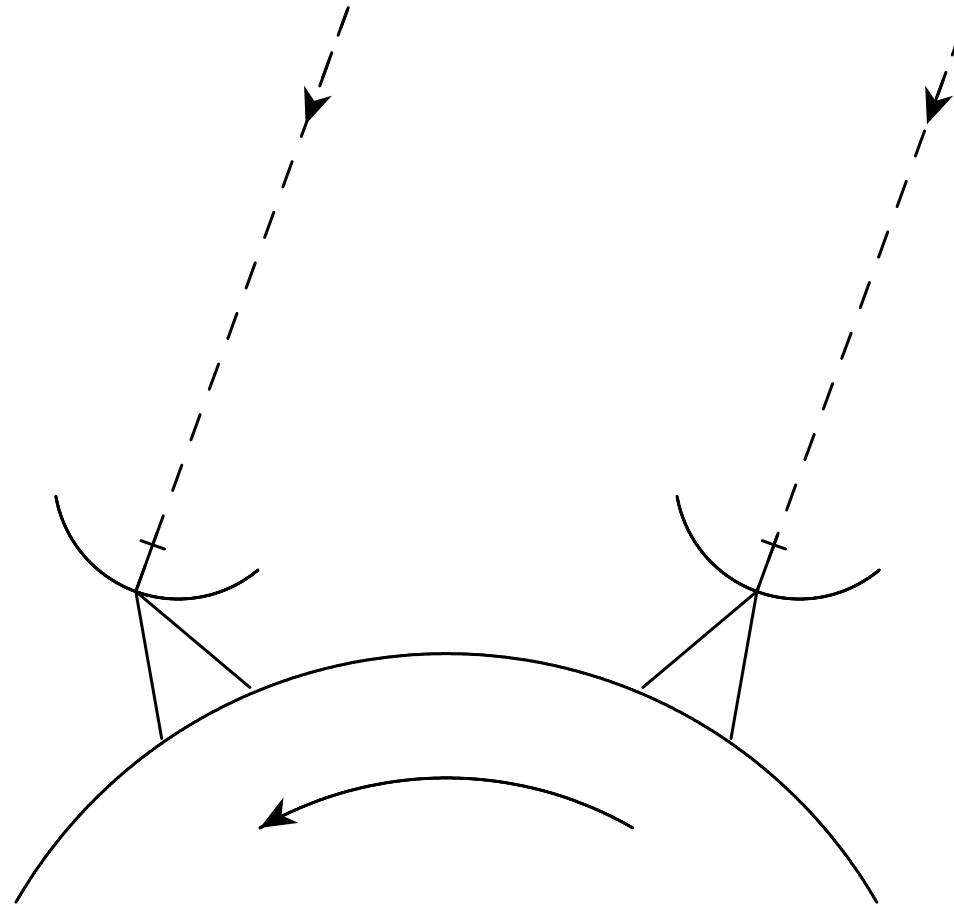
Interferometer vs. Young's Exp.

1. Young's slits = telescopes/antennas
2. Signal received by telescopes are combined by pairs
3. Fringe visibilities (amplitude and phase) are measured

⇒ Antenna don't move during the observation. One Fourier component of the source (i.e., one visibility) measured by one baseline (or antenna pair).

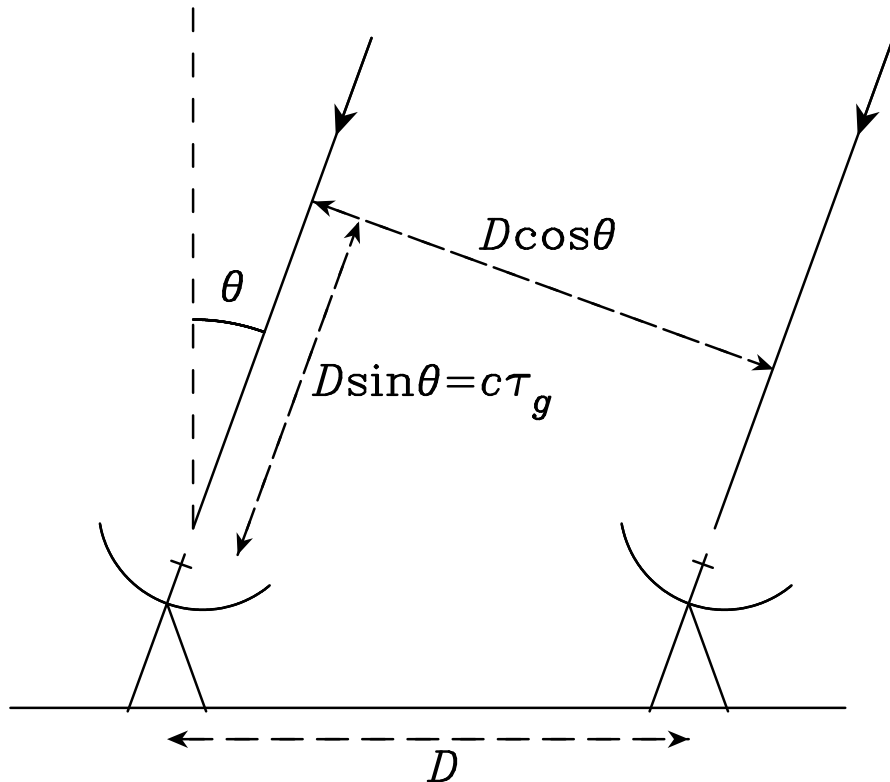
⇒ Each baseline length b_{ij} = a spatial frequency, measured in meter. Note that the projected length will change as the earth rotates.

Radio Interferometer



Elementary interferometer. The arrow indicates the rotation of the earth.

Radio Interferometer

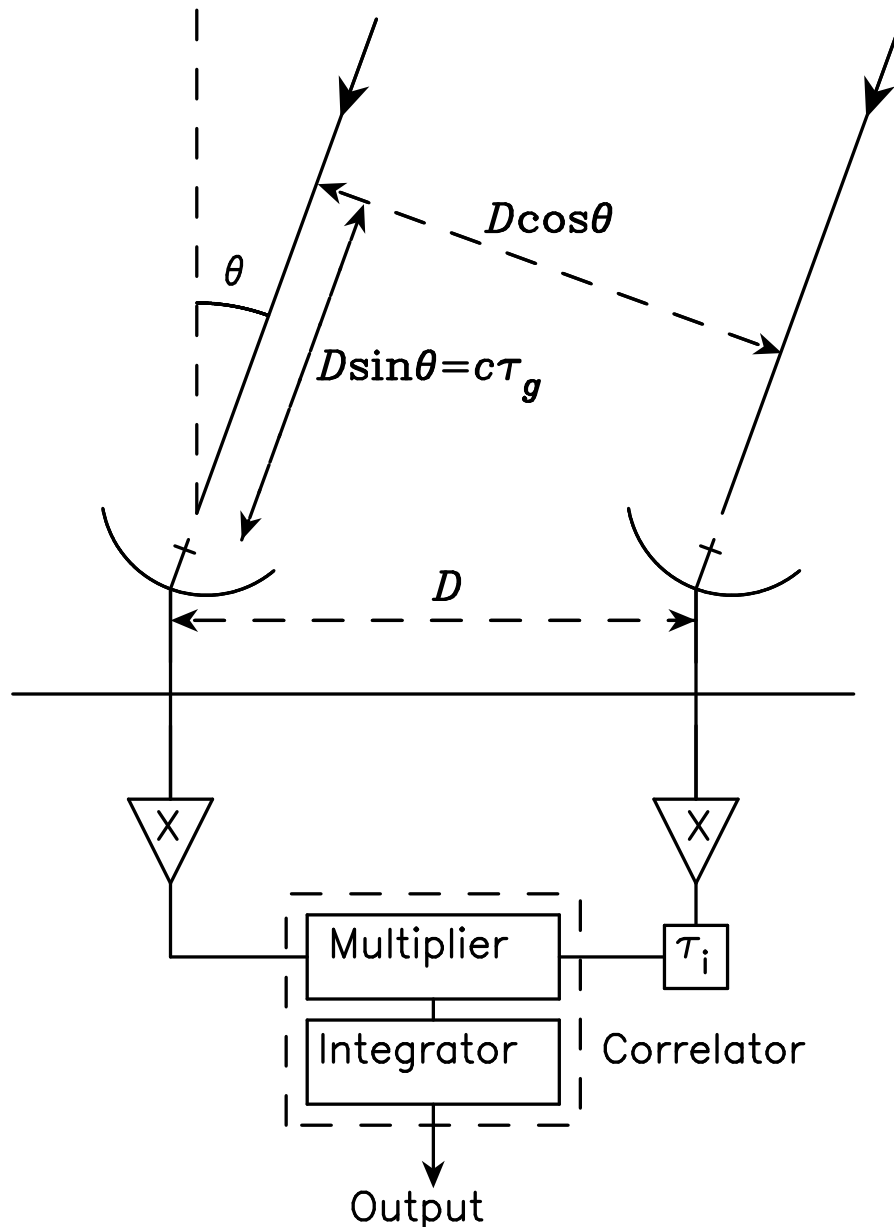


- D : baseline length
- θ : angle of the pointing direction from the zenith, changing with earth rotation.
- τ_g : geometrical delay $= \frac{D}{c} \sin \theta$

The wavefront from the source in direction θ is essentially **plane** because of great distance traveled, and it reaches the right-hand antenna at a time τ_g before it reaches the left-hand one:

Right: $E \cos(2\pi\nu(t - \tau_g))$ Left: $E \cos(2\pi\nu t)$

Radio Interferometer



- τ_g : Geometrical time delay
- τ_i : Instrumental time delay
- X: Bandpass amplifiers.
- Correlator: Multiplier + Integrator
- Output: Fringe Visibility

Multiplier:

Multiply the signals from the two antennas.

Integrator: Time Averaging Circuit, e.g., after integrating for 30 second in a SMA observation.

Radio Interferometer

Without any instrumental delay, the output of the multiplier is proportional to

$$\begin{aligned} F &= 2 \cos[2\pi\nu t] \cos[2\pi\nu(t - \tau_g)] \\ &= \cos(2\pi\nu\tau_g) + \cos(4\pi\nu t - 2\pi\nu\tau_g) \end{aligned}$$

with $\tau_g = \frac{D}{c} \sin \theta$. Here $2 \cos A \cos(A - B) = \cos B + \cos(2A - B)$. As the earth rotates,

$$\frac{d\tau_g}{dt} = \frac{D}{c} \cos \theta \frac{d\theta}{dt}$$

The most rapid rate of variation of θ is

$$\left. \frac{d\theta}{dt} \right|_{\max} = \Omega_E = 360^\circ/\text{day} \sim 7.25 \times 10^{-5} \text{ rad sec}^{-1} \sim 15 \text{ arcsec sec}^{-1}$$

where Ω_E is the [earth's rotational velocity](#). On the ground, the max. baseline given by the Very Long Baseline Array (VLBA) is $D \sim 10^4 \text{ km}$ (or 10^7 m), thus

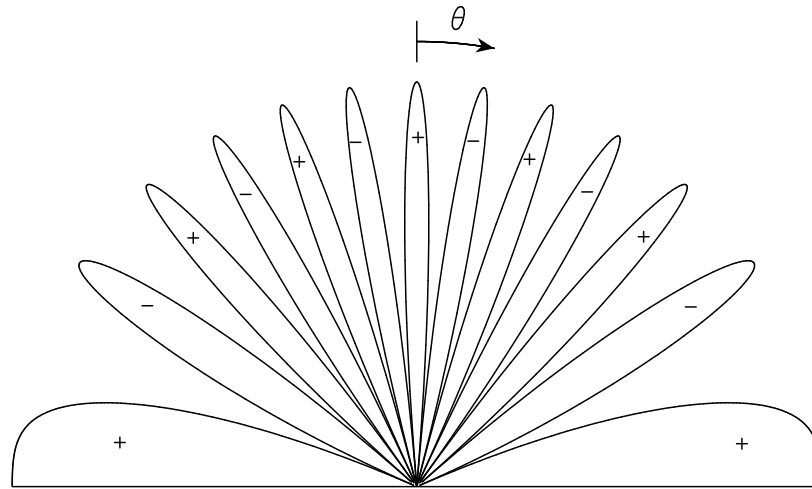
$$\frac{d\tau_g}{dt} < 2.4 \times 10^{-6} \quad \Rightarrow \quad \Delta\tau_g < 2.4 \times 10^{-6} \Delta t$$

Radio Interferometer

With time-averaging integrator, the more rapidly varying terms in F are easily filtered out leaving

$$r \equiv \frac{1}{T} \int_0^T F dt = \cos(2\pi\nu\tau_g) = \cos\left(\frac{2\pi D\xi}{\lambda}\right)$$

with $\xi = \sin \theta$. For sidereal sources the variation of θ with time as the earth rotates generates quasi-sinusoidal fringes at the correlator due to the variation of τ_g or explicitly the phase difference.



Polar plot of the fringe function r for $-90^\circ \leq \theta \leq 90^\circ$ (i.e, the whole sky) with $D/\lambda = 3$.

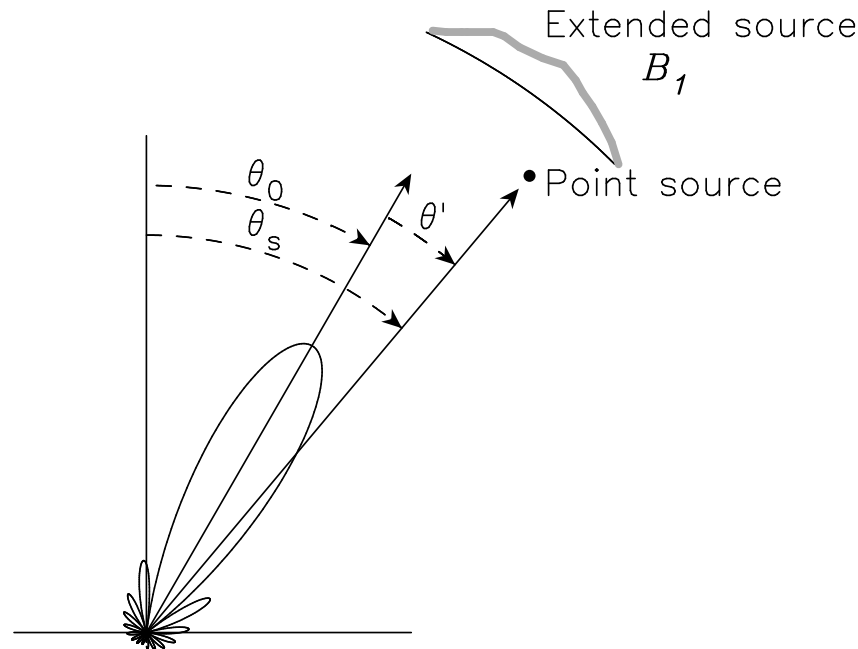
Nuisance!! Fringe frequency (cycle/sec or Hz) is $\nu d\tau_g/dt = D/\lambda \cos d\theta/dt$. It is ~ 7.5 Hz for the SMA with $D/\lambda \sim 10^5$, requiring unnecessary fast logging of data (i.e. 100 times/sec)!!! In addition, it is better to measure fringes in the vicinity of the maximum of the pattern $\Rightarrow$ [introducing \$\tau_i\$](#)

Radio Interferometer

Introducing a delay τ_i with $\tau = \tau_g - \tau_i$, the response in the output of the correlator is

$$r = \cos(2\pi\nu\tau)$$

Now let θ_0 be the pointing direction of the antenna. Consider a point source at position $\theta_s = \theta_0 + \theta'$, where θ' is very small.



Then the response is

$$r = \cos\left\{2\pi\nu\left[\frac{D}{c}(\sin\theta_0 + \theta' \cos\theta_0) - \tau_i\right]\right\}$$

Radio Interferometer

From the previous slide, we have

$$r = \cos\left\{2\pi\nu\left[\frac{D}{c}(\sin\theta_0 + \theta' \cos\theta_0) - \tau_i\right]\right\}$$

Setting $\tau_i = \tau_g(\theta_0) = (D/c) \sin\theta_0$, we have

$$r = \cos\left(2\pi\nu\frac{D}{c}\theta' \cos\theta_0\right) \equiv \cos(2\pi u\xi')$$

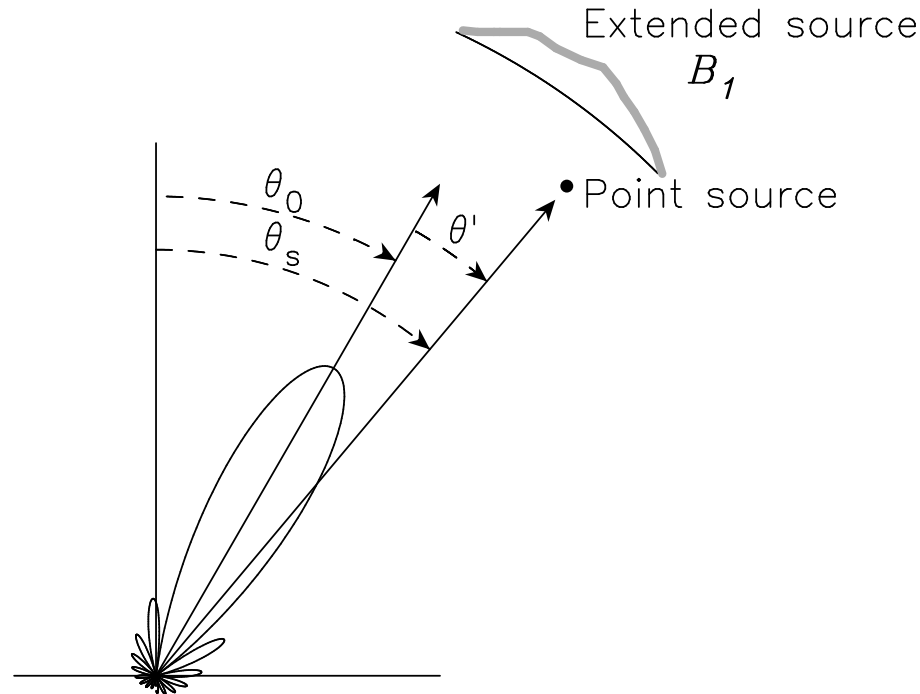
with $\xi' = \sin\theta' \approx \theta'$ and

$$u = \frac{D\nu}{c} \cos\theta_0 = \frac{D}{\lambda} \cos\theta_0$$

which is defined as the **spatial frequency** given by **the projected baseline in the unit of λ** . The spatial frequency can be measured in cycles/radian since ξ' is small and can be measured in radians. As the earth rotates, the response to a point source at the phase center θ_0 remains constant at the maximum value (i.e, 1) $\Rightarrow$ **Fringe Stopping for the pointing center**. Measured w.r.t. θ_0 , **r is the response of the correlator centered at the pointing center measured at different u** . This is what we need as **the antenna generally track the source under observation**. This sinusoidal fringe pattern is similar to that of the Young's double slits interference.

Single Antenna

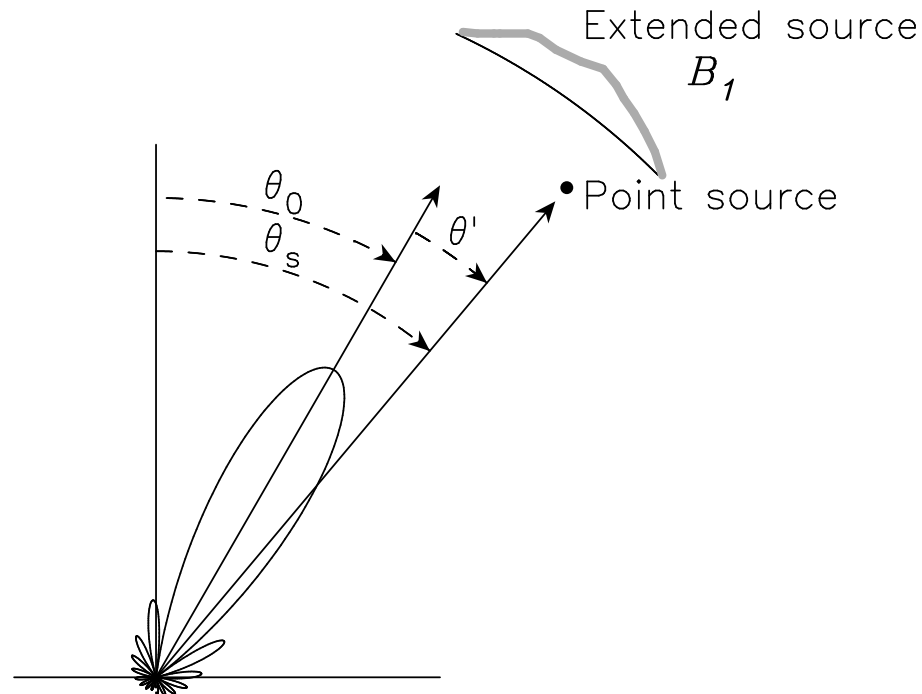
Consider the following 1D example for one single antenna:



Suppose that the pattern maximum is in direction θ_0 (pointing direction) and the point source direction is θ_s ; then the effective area for the direction θ_s is $A_1(\theta_s - \theta_0, \nu) \equiv A_e P_n$. Then the output power of the antenna for the point source is (one polarization):

$$\frac{1}{2} S(\nu) \Delta \nu A_1(\theta_s - \theta_0, \nu) \quad (\text{erg/s})$$

Single Antenna



For an extended source of brightness $B_1(\theta_s, \nu)$, the total output power from the antenna at pointing direction θ_0 is (one polarization):

$$\frac{\Delta\nu}{2} \int_{\text{source}} B_1(\theta_s, \nu) A_1(\theta_s - \theta_0, \nu) d\theta_s \quad (\text{erg/s})$$

$\Rightarrow$ cross correlation of the antenna reception pattern and the brightness distribution of the source.

Single Antenna

From previous slide, we have

$$\frac{\Delta\nu}{2} \int_{\text{source}} B_1(\theta_s, \nu) A_1(\theta_s - \theta_0, \nu) d\theta_s$$

Defining $\mathcal{A}_1(\theta_0 - \theta_s) = A_1(\theta_s - \theta_0)$, i.e., $\mathcal{A}_1$ is mirror image of A_1 , then

$$\frac{\Delta\nu}{2} \int_{\text{source}} B_1(\theta_s, \nu) \mathcal{A}_1(\theta_0 - \theta_s, \nu) d\theta_s$$

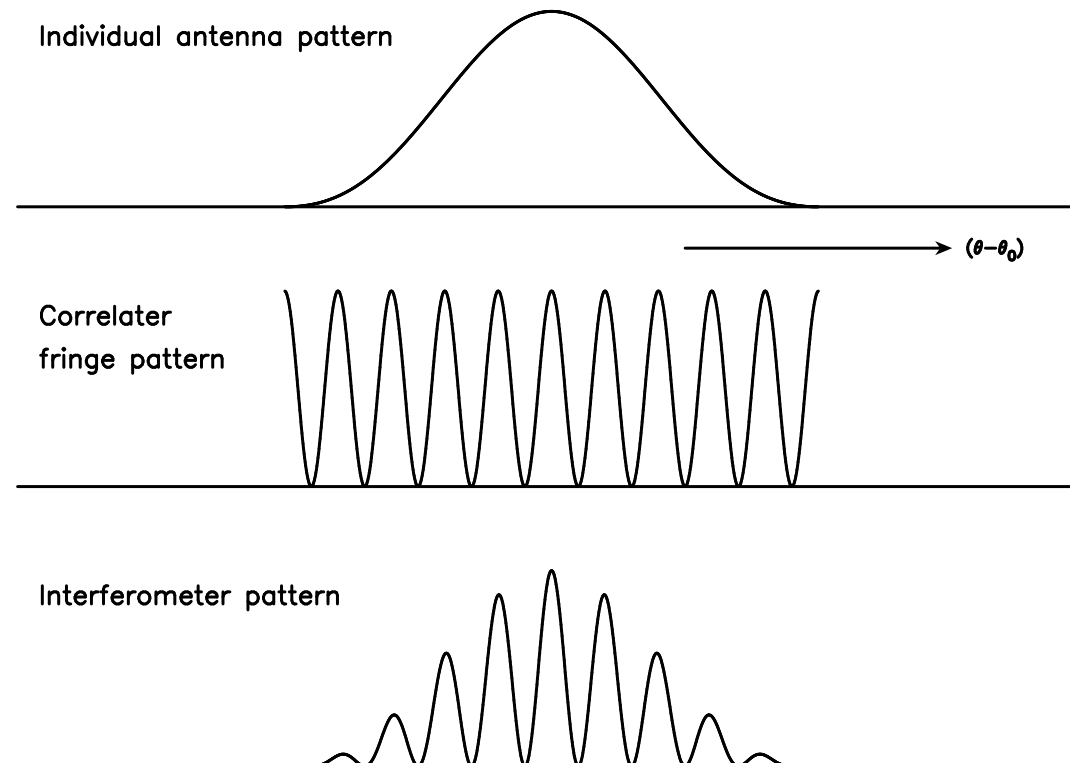
$\Rightarrow$ convolution of the source with the mirror image of the antenna reception pattern. Note that for most cases, since A_1 is symmetric about θ_0 , we have $\mathcal{A}_1 = A_1$.

Interferometer

The power reception pattern of an interferometer is the antenna reception power pattern multiplied by the correlator fringe pattern r . Thus, the power pattern with fringe stopping is given by

$$r_i(u, \theta) = \cos[2\pi u(\theta - \theta_0)] A_1(\theta - \theta_0)$$

which is a symmetric function about θ_0 . A_1 is renamed **primary beam pattern** of the antenna.



Interferometer

For an extended source of brightness $B_1(\theta_s)$, the output power of the interferometer is (one polarization) (omitting constant gain factors):

$$R(u) = \frac{\Delta\nu}{2} \int_{\text{source}} B_1(\theta_s) \cos[2\pi u(\theta_s - \theta_0)] A_1(\theta_s - \theta_0) d\theta_s$$

Since $\cos[2\pi u(\theta_s - \theta_0)] = \cos[2\pi u(\theta_0 - \theta_s)]$, we have a convolution

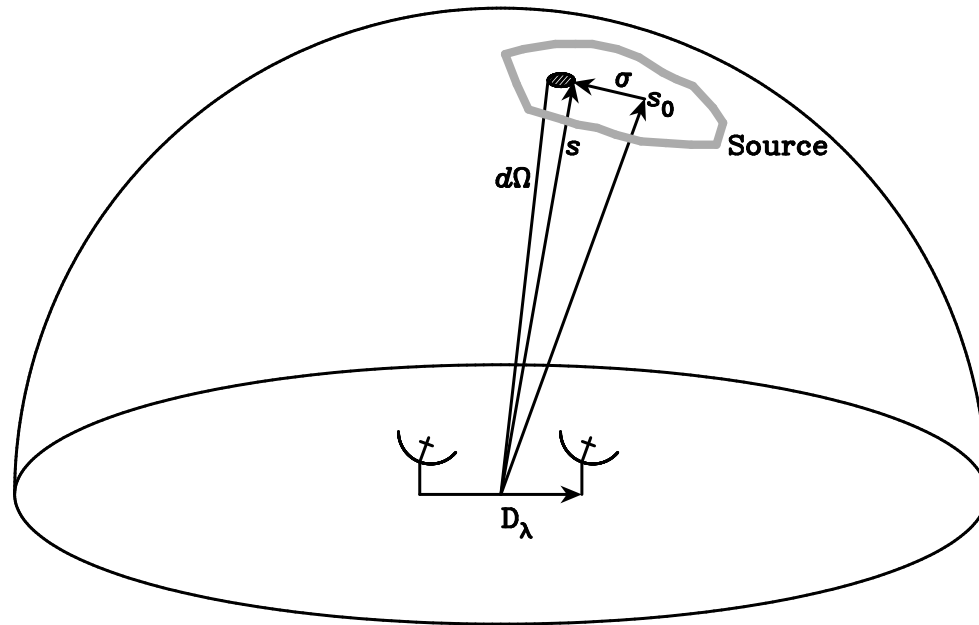
$$R(u) = \frac{\Delta\nu}{2} \int_{\text{source}} B_1(\theta_s) \cos[2\pi u(\theta_0 - \theta_s)] \mathcal{A}_1(\theta_0 - \theta_s) d\theta_s$$

Since the antennas always track the source, we can define $\xi' = \theta_s - \theta_0$ and $B_1(\theta_s) \rightarrow B(\xi')$. So,

$$R(u) = \frac{\Delta\nu}{2} \int_{\text{source}} B(\xi') \cos(2\pi u\xi') A_1(\xi') d\xi'$$

Radio Interferometer

Consider an interferometer tracking a source with the pointing center at $\mathbf{s}_0$.



Here $\mathbf{s}_0$, $\mathbf{s}$ and $\boldsymbol{\sigma}$ are all unit vectors with $\mathbf{s} = \mathbf{s}_0 + \boldsymbol{\sigma}$ and $\mathbf{D}_\lambda = \mathbf{D}/\lambda$. The geometrical delay is

$$\tau_g = \frac{\mathbf{D}_\lambda \cdot \mathbf{s}}{\nu} = \frac{\mathbf{D}_\lambda \cdot (\mathbf{s}_0 + \boldsymbol{\sigma})}{\nu}$$

where $\mathbf{D}_\lambda \cdot \mathbf{s}$ is the baseline length projected onto the $\mathbf{s}$ direction and $\nu = c/\lambda$.

Radio Interferometer

The output from the correlator is then proportional to

$$\begin{aligned} R(\mathbf{D}_\lambda, \mathbf{s}_0) &= \Delta\nu \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \cos[2\pi \mathbf{D}_\lambda \cdot (\mathbf{s}_0 + \boldsymbol{\sigma}) - 2\pi\nu\tau_i] d\Omega \\ &= \Delta\nu \cos(2\pi \mathbf{D}_\lambda \cdot \mathbf{s}_0 - 2\pi\nu\tau_i) \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \cos(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \\ &\quad - \Delta\nu \sin(2\pi \mathbf{D}_\lambda \cdot \mathbf{s}_0 - 2\pi\nu\tau_i) \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \sin(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \end{aligned}$$

Now we define the complex visibility as a Fourier Transform of $A(\boldsymbol{\sigma})B(\boldsymbol{\sigma})$:

$$\mathcal{V} = |\mathcal{V}| e^{i\phi_v} = \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) e^{-i2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}} d\Omega$$

Expanding both sides, we have

$$\begin{aligned} \mathcal{V} = |\mathcal{V}| \cos \phi_v + i |\mathcal{V}| \sin \phi_v &= \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \cos(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \\ &\quad - i A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \sin(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \end{aligned}$$

Radio Interferometer

Then by separating the real and imaginary parts we obtain

$$\begin{aligned}\int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \cos(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega &= |\mathcal{V}| \cos \phi_v \\ \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) \sin(2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega &= -|\mathcal{V}| \sin \phi_v\end{aligned}$$

Then the output from the correlator is

$$\begin{aligned}R(\mathbf{D}_\lambda, \mathbf{s}_0) &= \Delta\nu \cos(2\pi \mathbf{D}_\lambda \cdot \mathbf{s}_0 - 2\pi\nu\tau_i) |\mathcal{V}| \cos \phi_v \\ &\quad + \Delta\nu \sin(2\pi \mathbf{D}_\lambda \cdot \mathbf{s}_0 - 2\pi\nu\tau_i) |\mathcal{V}| \sin \phi_v\end{aligned}$$

and thus

$$R(\mathbf{D}_\lambda, \mathbf{s}_0) = \Delta\nu |\mathcal{V}| \cos(2\pi \mathbf{D}_\lambda \cdot \mathbf{s}_0 - 2\pi\nu\tau_i - \phi_v)$$

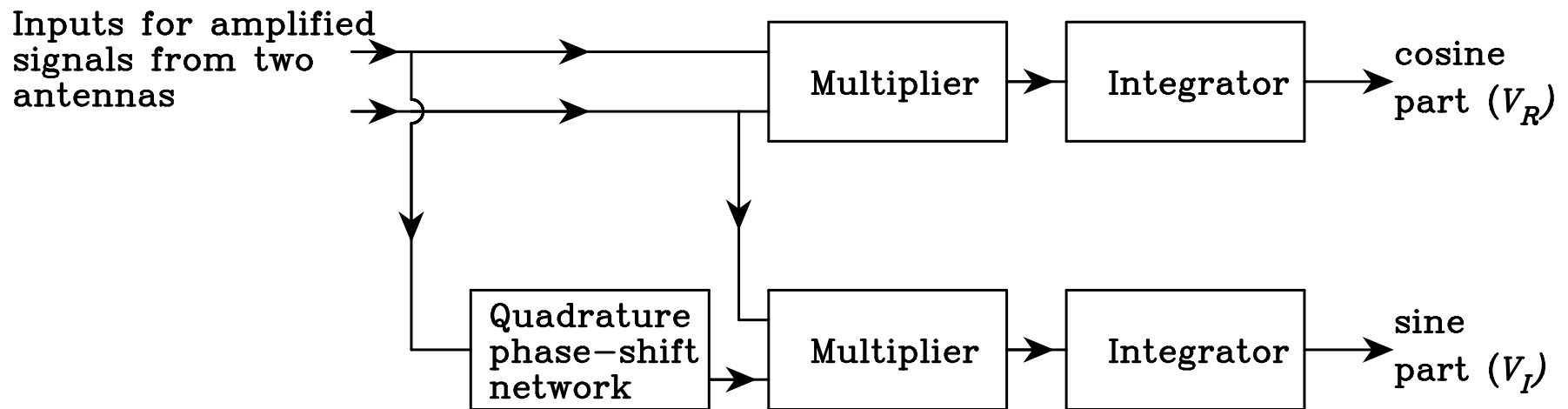
Note that if we set $\tau_i = \tau_g$ then the sine (imaginary) term will vanish, i.e,

$$R(\mathbf{D}_\lambda, \mathbf{s}_0) = \Delta\nu |\mathcal{V}| \cos \phi_v$$

However, with one measured quantity, we can't derive $|\mathcal{V}|$ and ϕ_v .

Radio Interferometer

To retrieve the sine (imaginary) term, we can add additional correlator with additional time delay of $\frac{1}{4\nu}$ (corresponding to a phase delay of $\frac{\pi}{2}$) so that $\tau_i = \tau_g - \frac{1}{4\nu}$ before the multiplier.



Use of two correlators to measure the real (cosine) and imaginary (sine) parts of the visibility. This device is called a complex correlator. Here, Quadrature means $\frac{\lambda}{4}$ or $\frac{\pi}{2}$. As a result, we redefine

$$R(\mathbf{D}_\lambda, \mathbf{s}_0) = \Delta\nu |\mathcal{V}| (\cos \phi_v + i \sin \phi_v) = \Delta\nu \mathcal{V}$$

with $\mathcal{V} = V_R + iV_I$, $\tan \phi_v = V_I/V_R$ and $|\mathcal{V}|^2 = V_R^2 + V_I^2$.

Radio v.s. optical Interferometer

	Optical	Radio
Detector: kind	Quadratic	Linear
Detector: Observable	$I = \langle E \cdot E^* \rangle$	$ E \exp(i\psi)$
Measure: method	Optical fringes	Electronic correlation
Measure: Quantity	$ C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$	$ \mathcal{V} \exp(i\phi_V)$
Interferometer kind	Additive	Multiplicative

Advantage of Multiplicative interferometer: [all offsets are irrelevant](#) $\Rightarrow$ Much easier.